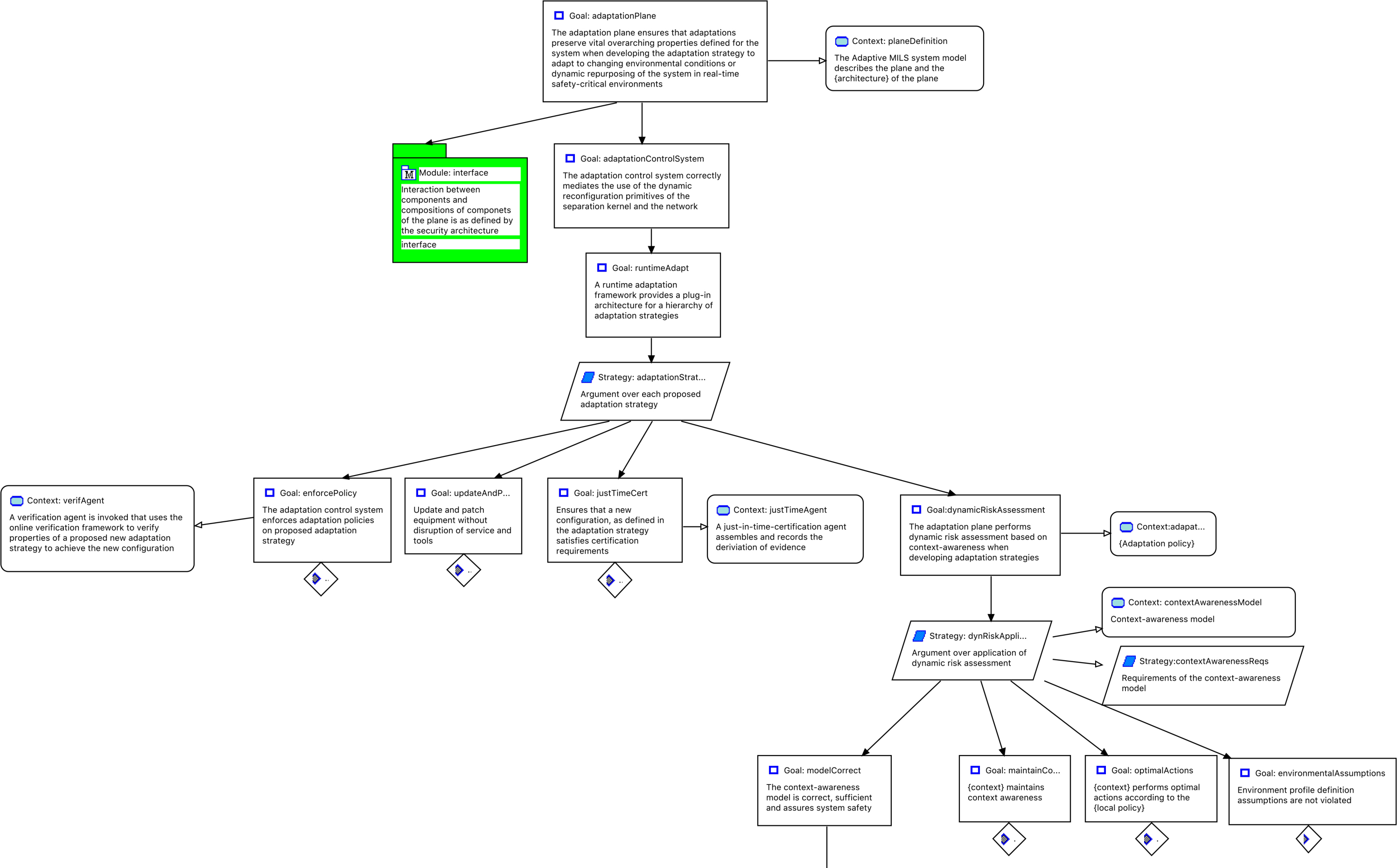
Goal: adaptationPlane
The adaptation plane ensures that adaptations preserve vital overarching properties defined for the system when developing the adaptation strategy to adapt to changing environmental conditions or dynamic repurposing of the system in real-time safety-critical environments
Context: planeDefinition
The Adaptive MILS system model describes the plane and the {architecture} of the plane
Module: interface
Interaction between components and compositions of componets of the plane is as defined by the security architecture
interface
Goal: adaptationControlSystem
The adaptation control system correctly mediates the use of the dynamic reconfiguration primitives of the separation kernel and the network
Goal: runtimeAdapt
A runtime adaptation framework provides a plug-in architecture for a hierarchy of adaptation strategies
Strategy: adaptationStrat...
Argument over each proposed adaptation strategy
Context: verifAgent
A verification agent is invoked that uses the online verification framework to verify properties of a proposed new adaptation strategy to achieve the new configuration
Goal: enforcePolicy
The adaptation control system enforces adaptation policies on proposed adaptation strategy
Goal: updateAndP...
Update and patch equipment without disruption of service and tools
Goal: justTimeCert
Ensures that a new configuration, as defined in the adaptation strategy satisfies certification requirements
Context: justTimeAgent
A just-in-time-certification agent assembles and records the deriviation of evidence
Goal:dynamicRiskAssessment
The adaptation plane performs dynamic risk assessment based on context-awareness when developing adaptation strategies
Context:adapat...
{Adaptation policy}
Context: contextAwarenessModel
Context-awareness model
Strategy: dynRiskAppli...
Argument over application of dynamic risk assessment
Strategy:contextAwarenessReqs
Requirements of the context-awareness model
Goal: modelCorrect
The context-awareness model is correct, sufficient and assures system safety
Goal: maintainCo...
{context} maintains context awareness
Goal: optimalActions
{context} performs optimal actions according to the {local policy}
Goal: environmentalAssumptions
Environment profile definition assumptions are not violated